UVTT v2 UPGRADER WORKSPACE

A Practical Blueprint and Structural Guide to Upgrading Legacy Maps

1. CORE RUNTIMES: WEB VS. DESKTOP PRO

While the standard Web App and the Desktop Pro application share the exact same user interface and drafting tools, their core architectures manage your computer's resources very differently. Choose the appropriate runtime based on your campaign's scale, map complexity, and available offline assets:

Feature Parameter	Web Application (Browser SPA)	Desktop Pro (Native Client)
File Ingestion Method	Manual drag-and-drop or file upload into browser window due to strict HTML5 security protocols.	Natively mounts your local hard drive OS folders; select your master asset folder once to link all contents instantly.
Streaming & RAM Scale	Constrained by strict browser-enforced memory caps. Best suited for quick edits or single-layer maps.	Natively streams gigabytes of high-resolution textures, tokens, and audio directly from your local drive.
Optimal Use-Case	Single-floor dungeons, quick touch-ups, or editing on-the-go on any machine.	Massive multi-story megadungeons, city maps, and heavy spatial audio soundscapes. Prevents memory crashes.

2. SCENIC TOUR OF THE WORKSPACE

The UVTT v2 Upgrader workspace is meticulously divided into five distinct zones to ensure that your drafting area remains clean and organized, allowing you to focus on the creative architectural process:

- **The Canvas Viewport (Center):** The central WebGPU-accelerated rendering canvas where your map artwork lives. Move your mouse around and check the *Toggle Grid Coordinate Cursor* button floating in the bottom-right corner to track your exact mathematical X/Y coordinates in real-time.
- **The Top Bar (Active Floor & History):** Anchored at the top-center. Houses the floor switcher (globe icon & dropdown), map layer renamer, Undo/Redo history states, and buttons to add blank levels, import legacy maps (.dd2vtt/.df2vtt), or delete active floors.
- **The Global Panel (Top Right):** The master control menu containing collapsible sub-panels for File & Export, Map Levels (Catalog), Format & Platform Exports, Action History (Undo lists), Environment Config, and Global Audio.
- **The Export Manager (Bottom Right):** Located beneath the Global Panel. Houses your active Export Profile dropdown and the prominent green **Run Compile & EXPORT** button.
- **The Left Drafting Toolbar:** Your main toolkit divided into four expandable categories: Architecture (structural tools), Entities (interactive objects), Assets (native folder explorer - Desktop Pro only), and Environment (visualization toggles).

3. THE STRUCTURAL TOOLKIT (ARCHITECTURAL DRAFTING)

Structural layout tools define the physical boundaries, lines of sight, and overhead ceilings of your map. Selecting an architectural tool with nothing selected configures its default properties, while selecting an existing line or node switches the panel to **Editing Mode** to expose clipboard utilities and conversion settings.

The Wall Tool (Geometry & Vision)

Walls shape character visibility, token collisions, and light casting. In Editing Mode, you can use standard clipboard actions (Copy, Paste, Duplicate, Delete) or toggle visibility overrides for the Player View.

- **Apply Smoothing Pass:** Mathematically curves linear walls using Bezier curves. *Note: This utility is only available if your selected wall has at least three vertices (two segments), as single straight lines cannot be curved.*
- **Collision Presets:** Select standard presets like *Solid* (blocks token movement and light/sight) or *Ethereal* (blocks tokens but allows light and sight—perfect for glass, energy barriers, or deep chasms).
- **Directional Blocking:** Configures one-way Line-of-Sight rules using the *Right-Hand Rule*. Vision is blocked from one side of your drawn segment while remaining completely transparent from the other side (ideal for cliff edges, elevated ledges, or arrow slits).
- **Z-Height Bounds:** Defines the exact bottom and top elevation of the wall in standard grid units, allowing you to create vertical levels, balconies, or half-height structures.
- **Snap All Vertices:** Instantly shifts the selected wall nodes (or locks future nodes) directly to the active grid intersections.

The Portal Tool (Doors, Windows, & Grates)

Portals define structural entryways that can change their physical state during gameplay. GMs can bulk-edit multiple selected portals to quickly align their properties:

- **Portal Architecture:** Define the physical type: *Door* (toggleable path), *Window* (blocks tokens but lets light/sight pass), *Secret Door* (invisible on the player VTT until discovered), or *Portcullis*.
- **Initial State:** Set the starting state for your session: *Open*, *Closed*, or *Locked*.
- **Z-Height Bounds:** Restricts the portals' vertical elevations, perfect for high prison bars, elevated arrow slits, or floor grates.

The Roof Tool (Canopies & Overhead Covers)

Roofs define solid overhead polygons representing forest canopies, ceilings, or upper-story floors. Compliant VTT engines will dynamically fade or completely hide these polygons when a player's token steps underneath them:

- **Roof Tint & Opacity:** Configures the visual overlay and transparency percentage for the GM view or on the map before tokens enter.
- **Z-Height Bounds:** Establishes upper and lower height bounds, enabling multi-tier maps where players can walk both on top of and beneath a roof structure.
- **Hidden from Players:** Keeps the overhead geometry completely invisible on the player side while keeping it fully operational for the GM.

Grid Alignment ('Rubber Sheeter')

Aligning legacy map artwork grids with the VTT's coordinate system is known as Rubber Sheeting. The Grid Alignment tool provides two distinct methods to achieve perfect coordinate calibration:

- **The Automated 3-Step Workflow:** (1) Click *Pin Origin* to mark the top-left intersection of a map grid square. (2) Adjust DPI inputs to scale the grid. (3) Or click *Auto-Align* and drag a green bounding box over a single 1x1 map square on the artwork; the system will automatically calculate the optimal DPI and pixel offsets.
- **Manual Micro-Nudging:** Input exact values into the *Grid X/Y (DPI)* and *X/Y Offset (px)* number boxes. Use the physical micro-nudge buttons (→ **Step +X**, ← **Step -X**, ↓ **Step +Y**, and ↑ **Step -Y**) to shift the grid pixel-by-pixel for flawless coverage.

4. THE INTERACTIVE TOOLKIT (DYNAMIC ENTITIES & AUTOMATION)

Interactive entities are placed elements that bring your digital environment to life. All interactive tools share a **Universal Editing State**—selecting any placed entity opens clipboard features (Copy, Paste, Duplicate, Delete), quick clone buttons, and player visibility overrides.

The Light & Spatial Audio Tools

Create dynamic lighting and highly localized auditory ambiances with physics-based falloff controls:

- **Dynamic Lighting:** Project light as an omnidirectional *Point* source or a focused *Cone/Beam*. Set the floating *Z-Axis Elevation* (grid units), dial in *Hex Colors & Intensity*, adjust *Bright/Dim Radii*, and select a *Decay Model* (Inverse Square vs. Linear). Apply organic movement with *Animation Profiles* (flicker, pulse, strobe).
- **Spatial Audio:** Loop a localized sound file. Adjust *Base Volume* and establish acoustic ranges via *Inner Core* (100% volume) and *Max Range* (fades to silence). Check **Muffled by Walls (Occlusion)** to tell the VTT to realistically attenuate sound if a solid wall breaks the player token's line of sight.

The Spawn, Emitter, & Event Tools

Control player placement, environmental weather, and mechanical scripting without writing a single line of code:

- **The Spawn Tool:** Place a player *Landing Zone*, choose its physical footprint on the grid, and use the *Token Arrival Heading* slider (degrees) to rotate their starting view. *Crucial: Every map layer must have exactly one designated Default Landing Zone to load players successfully.*
- **The Emitter Tool:** Add atmospheric particle effects (Rain, Snow, Embers, Fog). Render them globally or locally in a single room. Adjust Z-Axis levels, layering relative to roofs, and use sliders for scale, direction, speed, density, and variance.
- **The Event Tool & Wire Targets:** Define a trigger zone (footprint), choose an *Activation Method* (On Step, On Click, GM Only), and select a *Target Action* (Teleport, Toggle Visibility, Play Sound). To build a trap: select both the Event Trigger and your Target Object (like a Door) on the canvas to open the **WIRE TARGETS** sub-panel. Click **Bind Selected Entities** to wire them, or **Clear Targets** to sever the link.

> ■ **ADVANCED WORKFLOW: MULTI-LEVEL TELEPORTATION & STAIRS**
> When setting up stairs or elevators, select the **Event Tool** and set the Event Type to **'Teleport'** or **'Stairs/Ladder'** to open the **Destination Routing** controls in the floating panel. Under Destination Routing:
> • **Target Map/Floor:** Select the target destination level from your project catalog (e.g., Basement).
> • **Target Landing Zone:** Choose the exact named player Spawn Point on that destination floor where tokens will arrive.
> • ■ **Automated Two-Way Linking (New Event Placements Only):** Check *'Create return Spawn point'* to automatically drop a linked player Landing Zone on the grid next to your starting stairs. Check *'Auto-Create Reciprocal Return Link'* to automatically generate the exact reverse stairs and spawn point on the destination floor, giving you a fully wired, two-way staircase with a single click!

5. DESKTOP PRO EXCLUSIVE: VTT SIMULATION MODE

In the standard Web version of the Upgrader, the *Player View* toggle acts as a structural testing utility to quickly check line-of-sight blocks and basic lighting layout.

The **Desktop Pro** version leverages its unrestricted WebGPU access and native Wails/Go backend to provide a fully interactive **VTT Simulation Mode** button right next to it on the Environment toolbar. Clicking this button transforms your raw drafting canvas into a fully functional, real-time WYSIWYG VTT engine. It is the ultimate tool for pre-production testing, letting you direct and experience your dungeon's atmosphere before exporting:

- **Dynamic Lighting Rendering:** The canvas instantly plunges into your configured ambient darkness level, illuminated strictly by your placed light sources. Verify your exact hex colors, brightness values, and decay models as they interact with walls.
- **Active Animations:** Experience the map in motion. Watch torches flicker, arcane runes pulse, and searchlights strobe at your exact configured speeds, variance settings, and angles.
- **Live Particle Emitters:** Local and global weather effects (rain, snow, ashes, fog) generate instantly. Because Desktop Pro is a native client, it streams these high-resolution particle textures directly from your local hard drive, providing a silky-smooth, crash-free preview of heavy visual environments without browser memory caps.

6. GLOBAL MAP LAYERS & CACHE MANAGEMENT

Configure underlying grid math and protect your computer's resources during long mapping sessions:

- **Environment Config:** Define your *Grid Topology* (Square, Hex Horizontal, Hex Vertical, Isometric), *Grid Size* (pixel-per-grid resolution, e.g., 70px or 100px), *Grid Color*, and separate line weight sliders for the *Main Grid* and *Subgrid* lines.
- **Global Audio:** Assign an overarching *Background Soundtrack* (e.g., exploration theme) and continuous *Ambient Soundscape* loops (e.g., whispering winds) that play map-wide regardless of camera focus.
- **Map Levels (The Catalog):** Manage multi-floor hierarchies in a structured list. Rename layers with the Pencil icon, or permanently purge them and their assets using the Trash Can icon.
- **■■ CRITICAL RAM SAVIOR (Action History Cache Control):** The application saves a complete snapshot of your map data with every single action to feed its robust undo/redo timeline. During long, detailed sessions, this state history will consume massive amounts of your web browser's RAM, which can result in sudden out-of-memory browser crashes. **GM Action:** Open the Global Panel's *Action History* and click the **Clear History** button. This flushes the undo cache, instantly freeing browser memory without affecting your current active map layout. Flush the cache every 45-60 minutes during heavy structural editing.

7. THE EXPORT & PACKAGING WORKFLOW

Translate your in-memory blueprint catalog into finalized, playable VTT assets:

- **Project State Management:** Use *Save Project* to download your raw project database (preserving original texture files, Bezier vectors, and event wires for future editing). Use *Load Project* to restore your work in a clean session.
- **Package Catalog as Compound Dungeon:** When checked, the compiler bundles all of your layers into a single **.uvtt2z** ZIP archive, automatically rewriting your teleportation/stairs events to link internally across floors. When unchecked, levels export in *Federated Mode* as completely independent files.
- **Format & Platform Compiling:** Choose your compile rules in the Global Panel: *Export v1 (Downgrade)* to roll back to standard .dd2vtt, *Compile v2 (Standard)*, *Secure Archive (.zip)* which generates cryptographically secure *manifest.hash* files to protect premium intellectual property, or specialized platform compilers for Foundry VTT, Roll20, or Fantasy Grounds.

- **The Export Manager (Bottom Right):** Set your profile in the dropdown and click the big green **Run Compile & EXPORT** button to assemble, package, sign, and download your finished playable campaign assets.